

```
....After swapping, the range is: [5, 6, 7]
Recursively calling quicksort on: [5] and [7]

quicksort() called on this range: [5]
.....The full list is: [0, 1, 2, 3, 4, 5, 6, 7]

quicksort() called on this range: [7]
.....The full list is: [0, 1, 2, 3, 4, 5, 6, 7]

Sorted: [0, 1, 2, 3, 4, 5, 6, 7]
```

Quicksort is a commonly used sorting algorithm because it is straightforward to implement and, well, quick. The other commonly used sorting algorithm, merge sort, is also fast and uses recursion. We cover it next.